

# Todd Takala

*Lead Data Scientist — Lead Machine Learning Engineer — AI/ML Engineer*

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## Summary of Qualifications

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Lead Data Scientist with a proven track record of delivering over \$1.3 billion in business value through machine learning and end-to-end data science solutions to solve complex challenges. Manage complete project lifecycle, including stakeholder engagement, KPI definition, development, and deployment.

- **Machine Learning & AI Leadership:** Spearheaded the creation of an ML/AI system to predict equipment repair times, resulting in an \$80 million reduction in warranty costs and a CEO Award. Designed and optimized winning regression and recommendation systems, achieving two-time champion status in the Amazon Machine Learning University competition.
- **MLOps & End-to-End Pipelines:** Engineered and deployed robust ETL/MLOps pipelines using Snowflake and AWS services (SageMaker, S3, Step Functions) to automate model retraining. Championed MLOps best practices, setting new standards for GitHub CI/CD, Docker, and AWS architecture. Developed an end-to-end MLOps pipeline to deploy AI solutions for analyzing IoT and telematics data.
- **Data Analysis & Engineering:** Resolved critical design flaws in mining equipment through reproducible analysis in R and SQL, saving clients \$1.2 billion over six years. Created a regression model that identified key performance factors, leading to \$62 million in savings within 90 days. Analyzed over 2 petabytes of IoT data from a 400,000-robot fleet to establish a new design-for-reliability benchmark.
- **Team Leadership & Technical Mentorship:** Directed a high-performance team of up to 8 engineers, fostering a data-driven approach to solve equipment performance issues. Provided technical leadership by developing foundational predictive forecasting models for supply chain management.

## Experience

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**Lead Data Scientist** — Caterpillar Inc. — Surprise, AZ September 2022 – Present

Led the end-to-end development of high-impact data science initiatives, including an award-winning ML system to predict equipment repair times. Engineered and deployed robust MLOps pipelines using AWS and Snowflake and established best practices for CI/CD and system architecture. Managed project lifecycles and created comprehensive monitoring plans to validate business impact.

**Reliability Analytics Manager** — Amazon — Tempe, AZ February 2022 – September 2022

Pioneered a new design-for-reliability benchmark by analyzing petabyte-scale IoT data from a large robotics fleet. Authored an influential white paper on complex Weibull analysis and secured adoption of the new benchmark by senior leadership.

**Data Engineering / Reliability Manager** — Empire-Cat — Mesa, AZ January 2021 – February 2022

Led a team to establish an MLOps pipeline with ARIMA and SVM for AI-driven forecasting in supply chain management, integrated data, managed a component exchange program, collaborated with clients on durability forecasting, and addressed design flaws in mining equipment, reducing client costs by \$1.2B. This involved creating a comprehensive machine learning dataset from NLP on work orders and IoT telemetry.

## Skills

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Python, Redshift, ECR, DBA, ETL, TensorFlow, PyTorch, Scikit-learn, Transformers, Anomaly Detection, Experiments, pandas, keras, numpy, Git, Strategic Thinking, Adaptability, Agile, Continuous Integration, Programming Languages, Distributed Computing, Architectural Design, Statistical Tools, Data Gathering, Willing to Learn, AI/ML, LLMs, Business Requirements, Large Language Models, Developer Tools, Growth Mindset, Creative, Researching, GitHub Copilot

## Education

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**Bachelor of Science in Mechanical Engineering** — Arizona State University — Tempe, AZ 2006  
STEM coursework included: Numerical Methods, Computational Thermodynamics, Statistics, Logic, Linear Algebra, Finite Element Analysis

## Certifications

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**Data Camp:** Data Scientist Professional Certificate, Certified SQL Associate, AI Fundamentals Certificate

**Canonical:** Ubuntu Linux Professional Certificate

**GitHub:** Career Essentials in GitHub Professional Certificate

**Udemy:** Streamlit for Snowflake Masterclass

**BerkeleyX:** The Science of Happiness at Work Professional Certificate

**Empire-Cat:** Lean 6 Sigma Green Belt

**Komatsu North America:** Certified Technical Communicator

**Society of Automotive Engineers:** Cost and Finance for Engineers

**Arizona State Board of Technical Registration:** EIT, Mechanical Engineering – Registration No. 10670